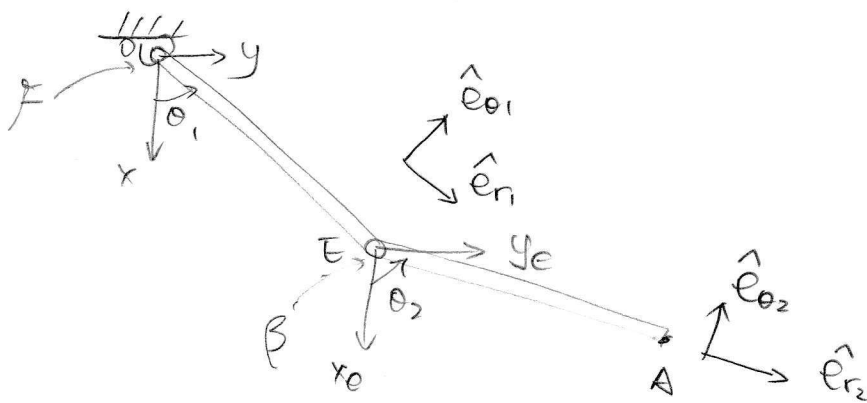


31. Double pendulum kinematics.

Consider a double pendulum where both links have length l .

Find the acceleration of the end of second link using two sets of minimal coordinates.

(a) Absolute angles. θ_1, θ_2 .



$$\vec{a}_{A/F} = \vec{a}_{A/\beta} + \vec{a}_{E/F}$$

$$\vec{a}_{E/F} = \underbrace{-\dot{\theta}_1^2 l \hat{e}_{r_1}}_{\text{centripetal}} + \underbrace{\ddot{\theta}_1 l \hat{e}_{\theta_1}}_{\text{tangential}}$$

Similarly,

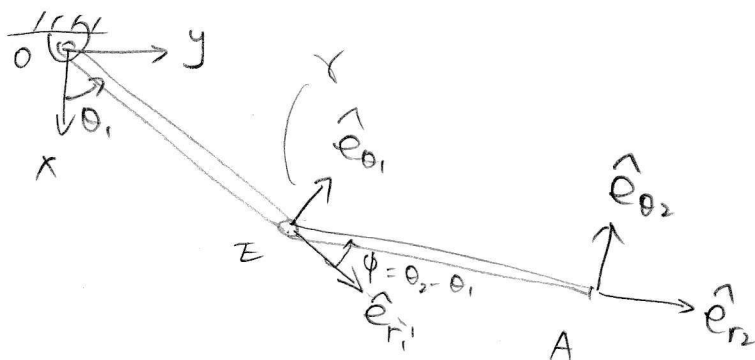
$$\vec{a}_{A/\beta} = -\dot{\theta}_2^2 l \hat{e}_{r_2} + \ddot{\theta}_2 l \hat{e}_{\theta_2}$$

Thus,

$$\boxed{\vec{a}_A = -\dot{\theta}_1^2 l \hat{e}_{r_1} + \ddot{\theta}_1 l \hat{e}_{\theta_1} - \dot{\theta}_2^2 l \hat{e}_{r_2} + \ddot{\theta}_2 l \hat{e}_{\theta_2}} \quad (1)$$

Solution with absolute angle.

(b) Relative angles θ_1 , $\phi = \theta_2 - \theta_1$



frame $\hat{e}_{r_1}, \hat{e}_{\theta_1}(\gamma)$
is attached at point E,
rotating with upper link

Since intermediate frame is rotating, use 5-term formulas

$$\vec{a}_{A/E} = \vec{a}_{E/\mu} + \vec{a}_{A/\gamma} + (-\omega^2 \vec{r}_{A/E}) + \vec{\alpha} \times \vec{r}_{A/E} + 2\vec{\omega} \times \vec{v}_{A/E}$$

where,

$$\vec{a}_{E/\mu} = \text{Same as part (a)} = -\dot{\theta}_1^2 l \hat{e}_{r_1} + \ddot{\theta}_1 l \hat{e}_{\theta_1}$$

$$\vec{a}_{A/\gamma} = -\dot{\phi}^2 l \hat{e}_{r_2} + \ddot{\phi} l \hat{e}_{\theta_2} \quad \text{similar as before, but now it's } \phi$$

$$\vec{r}_{A/E} = l \hat{e}_{r_2}, \quad \omega = \dot{\theta}_1, \quad \alpha = \ddot{\theta}_1$$

$$\vec{\alpha} \times \vec{r}_{A/E} = \dot{\theta}_1 \hat{k} \times l \hat{e}_{r_2} = \dot{\theta}_1 l \hat{e}_{\theta_2}$$

$$\vec{v}_{A/E} = \dot{\phi} l \hat{e}_{\theta_2}, \quad \vec{\omega} \times \vec{v}_{A/E} = \dot{\theta}_1 \hat{k} \times \dot{\phi} l \hat{e}_{\theta_2} = -\dot{\phi} l \dot{\theta}_1 \hat{e}_{r_2}$$

Then,

$$\begin{aligned} \vec{a}_{A/E} = & (-\dot{\theta}_1^2 l \hat{e}_{r_1} + \ddot{\theta}_1 l \hat{e}_{\theta_1}) + (-\dot{\phi}^2 l \hat{e}_{r_2} + \ddot{\phi} l \hat{e}_{\theta_2}) \\ & - \dot{\theta}_1^2 l \hat{e}_{r_2} + \ddot{\theta}_1 l \hat{e}_{\theta_2} - 2\dot{\phi} l \dot{\theta}_1 \hat{e}_{r_2} \end{aligned}$$

collect terms.

$$\begin{aligned} \vec{a}_{A/E} = & -\dot{\theta}_1^2 l \hat{e}_{r_1} + \ddot{\theta}_1 l \hat{e}_{\theta_1} \\ & + \underbrace{(-\dot{\phi}^2 - \dot{\theta}_1^2 - 2\dot{\phi}\dot{\theta}_1)}_{-(\dot{\phi} + \dot{\theta}_1)^2} l \hat{e}_{r_2} + (\ddot{\phi} + \ddot{\theta}_1) l \hat{e}_{\theta_2} \end{aligned}$$

Then,

$$\vec{a}_{A/E} = -\ddot{\theta}_1^2 l \hat{e}_{r_1} + \ddot{\theta}_1 l \hat{e}_{\theta_1} - (\dot{\phi} + \dot{\theta}_1)^2 l \hat{e}_{r_2} + (\ddot{\phi} + \ddot{\theta}_1) l \hat{e}_{\theta_2} \quad (2)$$

↑↑
Solution with relative angle

Now, let's show (1) = (2)

Obviously, since $\phi = \theta_2 - \theta_1$,

$$\dot{\phi} = \dot{\theta}_2 - \dot{\theta}_1, \quad \ddot{\phi} = \ddot{\theta}_2 - \ddot{\theta}_1$$

$$\Rightarrow \dot{\theta}_2 = \dot{\phi} + \dot{\theta}_1, \quad \ddot{\theta}_2 = \ddot{\phi} + \ddot{\theta}_1 \rightarrow (2)$$

we get

$$\vec{a}_{A/E} = -\dot{\theta}_1^2 l \hat{e}_{r_1} + \ddot{\theta}_1 l \hat{e}_{\theta_1} - \dot{\theta}_2^2 l \hat{e}_{r_2} + \ddot{\theta}_2 l \hat{e}_{\theta_2}$$

which is same as (1).